

Chapter 2: Prompt Engineering Basics

Class 8 – Writing Effective Text Prompts

Learning Objectives

After completing this lesson, students will be able to:

- Understand the characteristics of an effective text prompt.
 - Write clear, specific, and well-structured prompts.
 - Use context to improve AI responses.
 - Apply role-based prompting techniques.
 - Identify common prompt writing mistakes.
 - Improve AI-generated responses by refining prompts.
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Introduction

Artificial Intelligence (AI) responds based on the instructions given by the user. A well-written prompt helps AI understand exactly what is required, resulting in more accurate, useful, and relevant answers.

An effective prompt is **clear, specific, and complete**. Instead of asking vague questions, students should provide enough details such as the topic, purpose, audience, format, and expected length. This lesson teaches students how to write prompts that produce better AI responses.

What is an Effective Text Prompt?

An **Effective Text Prompt** is a clear and detailed instruction that helps an AI system understand exactly what the user wants.

A good prompt should tell the AI:

- What to do
- What topic to cover
- Who the audience is
- What format to use
- How detailed the answer should be

Characteristics of an Effective Prompt

A good prompt should be:

- Clear
 - Specific
 - Simple
 - Complete
 - Relevant
 - Well-organized
 - Easy to understand
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Writing Clear and Specific Instructions

Clear instructions help AI generate accurate responses.

Weak Prompt

Tell me about animals.

Improved Prompt

Explain five characteristics of mammals in simple English for Class IX students using bullet points.

The improved prompt gives the AI:

- Topic
 - Number of points
 - Language level
 - Output format
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Using Context in AI Prompts

Context gives AI additional information about what you need.

Example Without Context

Write about pollution.

Example With Context

Write a 150-word article about air pollution for a Class IX science assignment using simple language.

Providing context improves the quality of the response.

Asking Open-ended and Closed Questions

Open-ended Prompt

Allows detailed responses.

Example:

Explain how renewable energy can reduce pollution.

Closed Prompt

Requires a short or direct answer.

Example:

Is solar energy a renewable source of energy?

Role-based Prompting

Role-based prompts ask AI to respond as a specific person or professional.

Examples

- Act as a science teacher and explain photosynthesis.
- Act as an English teacher and correct this paragraph.
- Act as a career counselor and suggest career options after Class X.
- Act as a graphic designer and suggest poster ideas.

Role-based prompts often produce more focused and useful responses.

Step-by-Step Prompt Writing

Students should follow these steps:

Step 1: Identify the topic.

Step 2: State the purpose.

Step 3: Add necessary context.

Step 4: Specify the output format.

Step 5: Mention the audience or class level.

Step 6: Review the prompt before submitting.

Prompt Formula

A simple formula for writing effective prompts:

Task + Topic + Context + Format + Audience + Length

Example

Explain the importance of water conservation in simple English using five bullet points for Class IX students.

Improving AI Responses

If the first response is not satisfactory, students can improve it by asking AI to:

- Simplify the language.
- Add more examples.
- Make it shorter.
- Explain step by step.
- Use bullet points.
- Include a table.
- Add a conclusion.

Common Prompt Writing Mistakes

Avoid these mistakes:

- Writing very short prompts.
 - Giving unclear instructions.
 - Asking multiple unrelated questions together.
 - Not mentioning the audience.
 - Forgetting to specify the output format.
 - Accepting the first response without reviewing it.
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Best Practices for Text Prompting

Students should:

- Use simple language.
 - Be specific.
 - Add relevant details.
 - Mention the desired format.
 - Review AI responses.
 - Improve prompts when necessary.
 - Verify important information.
 - Use AI responsibly.
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Examples of Good Prompts

Education

Explain Newton's Laws of Motion with simple examples for Class IX students.

Research

List five advantages and five disadvantages of online learning in a table.

Writing

Write a 200-word essay on Environmental Protection using simple English.

Presentation

Create a five-slide presentation outline on Artificial Intelligence.

Revision

Prepare ten revision questions from the chapter on Climate Change.

Real-Life Applications

Effective text prompts are used in:

- School assignments
 - Research
 - Presentation preparation
 - Essay writing
 - Project work
 - Coding assistance
 - Business communication
 - Content creation
 - Customer support
 - Professional documentation
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Career Opportunities

Prompt writing skills are valuable for careers such as:

- Prompt Engineer
- AI Trainer
- Content Writer
- Teacher
- Technical Writer

- Research Analyst
 - Software Developer
 - Digital Marketing Specialist
 - UX Writer
 - Business Consultant
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Revision

Students should revise:

- Effective Text Prompts
 - Characteristics of Good Prompts
 - Prompt Formula
 - Role-based Prompting
 - Context in Prompts
 - Prompt Improvement
 - Common Mistakes
 - Best Practices
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Did You Know?

- AI often provides better responses when prompts include the audience, format, and level of detail.
 - Small changes to a prompt can significantly improve the quality of AI-generated content.
 - Many professionals refine the same prompt several times before achieving the desired result.
 - Effective prompt writing is becoming an essential digital skill across many industries.
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Points to Remember

- A good prompt is clear, specific, and complete.
- Adding context improves AI responses.
- Role-based prompts provide more focused answers.
- Review and refine prompts when needed.
- Always verify important AI-generated information before using it.